

Soumyaratna Debnath

PhD Candidate and NPGS Scholar

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Research Interests

Embodied Spatial Intelligence | Neuroscience-Inspired Adaptive Perception | Omnidirectional Scene Understanding | Vision-Language-Action Learning | 3D Shape Analysis

Educational Qualification

DEGREE	INSTITUTION	CPI / CGPA	YEAR
PhD (EEE)	NTU Singapore	4.5 / 5	2025-2029
M.Tech (CSE)	IIT Gandhinagar, Gandhinagar, India	9.67 / 10	2022-2025
B.Tech (CSE)	CGEC, West Bengal, India	9.34 / 10	2018-2022

Experiences

- Researcher at CVIG Lab, IIT Gandhinagar** Mar 2024 – Apr 2025
Served as a *Project Fellow* at the Computer Vision, Imaging, and Graphics (CVIG) Lab, IIT Gandhinagar. I collaborated with the National Institute of Mental Health (NIMH), USA, and contributed to projects in partnership with L&T Technology Services, the AI Centre of Excellence (AICoE), the Ministry of Education, Government of India, and the Electronics Corporation of India Limited (ECIL).
- Software Developer at IMS, IIT Gandhinagar** Aug 2022 – Feb 2024
Developed and maintained modules of the Institute Management System (IMS), an ERP solution for IIT Gandhinagar, to streamline operations across departments and facilities.
- Research Intern, Department of CSE, CGEC, West Bengal** Sept 2020 – Feb 2022
Conducted research on soft computing and meta-heuristics-based algorithms for multi-level image segmentation and thresholding. These works were concluded in two conference publications and a book chapter.

Publications

- Seeing Only What Matters: Training-Free Adaptive Perception for Vision-Language Models in Dynamic Scenes**
Manuscript Under Review
Soumyaratna Debnath, Zinan Liu, and Lin Wang
- LLMind: Bio-inspired Training-free Adaptive Visual Representations for Vision-Language Models**
Conference on Computer Vision and Pattern Recognition (CVPR) 2026
Soumyaratna Debnath, Bui Duc Man, Zinan Liu and Lin Wang
- Mind Meets Space: Rethinking Agentic Spatial Intelligence from a Neuroscience-Inspired Perspective**
Manuscript Under Review
Bui Duc Manh, Soumyaratna Debnath, Zetong Zhang, Shriram Damodaran, Arvind Kumar, Yueyi Zhang, Lu Mi, Erik Cambria, and Lin Wang
- RASP: Revisiting 3D Anamorphic Art for Shadow-Guided Packing of Irregular Objects**
Conference on Computer Vision and Pattern Recognition (CVPR) 2025
Soumyaratna Debnath, Ashish Tiwari, Kaustubh Sadekar and Shanmuganathan Raman
- ScribGen: Generating Scribble Art through Metaheuristics**
ACM SIGGRAPH Asia 2024 Art Papers
Soumyaratna Debnath, Ashish Tiwari and Shanmuganathan Raman

6. **L3D-Pose: Lifting Pose for 3D Avatars from a Single Camera in the Wild**
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2025
Soumyaratna Debnath, Harish Khatti, Shashikant Verma and Shanmuganathan Raman
7. **STEP: Simultaneous Tracking and Estimation of Pose for Animals and Humans**
Manuscript under review for the International Journal of Computer Vision (IJCV).
Shashikant Verma, Harish Khatti, **Soumyaratna Debnath**, Yamuna Swami and Shanmuganathan Raman
8. **Modified Harris Hawk Optimization Algorithm for Multi-level Image Thresholding**
Published as a Chapter in Hybrid Computational Intelligent Systems
Soumyaratna Debnath, Abhirup Deb, Sourav De, Sandip Dey

Projects

1. **Shadow Guided Optimal Packing and Assembly of General 3D Shapes.** Developed a differentiable rendering based novel approach for optimizing the packing of arbitrarily shaped objects by analyzing shadows cast from different angles and extending this technique to guide the precise assembly of broken objects.
2. **AI-powered Traffic Management Solutions for Sustainable Cities.** Worked on an AI-powered traffic management solution, *IndAI Nagar*, to monitor urban traffic using CCTV in Indian settings. The project was undertaken by the AI Centre of Excellence (AICoE) under the Ministry of Education, Government of India.
3. **Analysis of Cotton Crop Fields using Aerial Imagery.** Leveraged drone-captured imagery to monitor and analyse cotton crops in agricultural fields. The initiative was carried out in collaboration with L&T Technology Services.
4. **Simultaneous Tracking and Estimation of Pose for Animals and Humans.** Conducted research to develop a novel transformer-based architecture for simultaneously tracking and estimating poses in primates for wildlife monitoring and behavioural studies. The research was in collaboration with the Laboratory of Neuropsychology (LN), National Institute of Mental Health (NIMH), United States.
5. **L3D-Pose.** The research involved lifting 2D poses to 3D using an Attention-driven MLP architecture and accurately retargeting them onto 3D avatars; in collaboration with NIMH, United States.

Achievements

1. Awarded the Nanyang President's Graduate Scholarship (NPGS) at NTU Singapore – the university's most prestigious scholarship, granted by the President of NTU to PhD students.
2. Awarded the Gold Medal for Outstanding Research in MTech – IIT Gandhinagar.
3. Offered the Director's Fellowship at IIT Gandhinagar – the most prestigious fellowship awarded to a master's student continuing into a PhD at the institute.
4. Qualified National Eligibility Test (NET) 2024 and GATE 2021, 2022 & 2025.
5. Finalist for the Qualcomm Innovation Fellowship India 2024-25 for the project proposal titled "Shadow Guided Optimal Packing and Assembly of General 3D Shapes".
6. Granted a Copyright from the Government of India for the software project "*Picrypt It*", which introduces an innovative method for embedding password-protected texts within colour images.
7. Awarded for Publishing in Top-Tier Conferences and Journals – IIT Gandhinagar
8. Awarded the Chairman Medal of Excellence for extraordinary academic performance from APS Binnaguri – 2018.
9. Awarded Certificate of Merit from the Central Board of Secondary Education for outstanding academic performance (a perfect 10 CGPA) in the All-India Secondary School Examination 2016.